CHAPTER 3

INTERIOR OF THE EARTH

CHAPTER ESSENCE AT A GLANCE:
Understand sources of internal Earth data, propagation of seismic P-waves and S-waves, the geometry of earthquake shadow zones, concentric layering of crust, mantle, and core, and volcanic intrusive landforms.

4 CORE PRINCIPLES TO UNDERSTAND FIRST

• **Sources of Interior Knowledge:** Direct sources include deep mining, borehole drilling (Kola Peninsula 12 km), and erupted volcanic lavas; indirect sources include gravitational anomalies, magnetic surveys, meteorites, and seismic wave behavior.

• **Seismic Wave Mechanics:** Body waves travel through the interior: P-waves (compressional, longitudinal, fastest, propagate through solids, liquids, and gases) and S-waves (transverse, shear, slower, propagate exclusively through solids).

• **Seismic Shadow Zones:** P-waves refract sharply at the mantle-core boundary creating a shadow band between 105 degrees and 145 degrees; S-waves cannot enter the liquid outer core, producing an extensive shadow zone beyond 105 degrees.

• **Concentric Chemical Layering:** Crust (oceanic basalt 5 km, continental granite 30 km; Moho discontinuity), Mantle (peridotite reaching 2900 km, containing asthenosphere between 80 and 200 km), and Core (liquid outer core, solid inner core composed of nickel and iron).

ESSENTIAL GEOGRAPHICAL MODELS & LANDFORMS

$$v_p = \sqrt{\frac{K + \frac{4}{3}\mu}{\rho}} \rightarrow \text{Primary Compressional Wave Velocity}$$

Seismic P-Wave Propagation Velocity in Elastic Media

$$v_s = \sqrt{\frac{\mu}{\rho}} \rightarrow \text{Secondary Shear Wave Velocity (Zero in Outer Core)}$$

Seismic S-Wave Velocity Demanding Rigidity Modulus

$$\text{Earth Core Density Contrast: } \rho_{\text{Crust}} \approx 2.7 \text{ to } 3.0 \text{ g cm}^{-3} \longleftrightarrow \rho_{\text{Core}} \approx 13.0 \text{ g cm}^{-3}$$

Radial Gravitational Density Contrast of the Earth

$$\text{Discontinuity Succession: Conrad} \rightarrow \text{Moho} \rightarrow \text{Repetti} \rightarrow \text{Gutenberg} \rightarrow \text{Lehmann}$$

Five Primary Internal Seismic Boundary Transitions

CRUCIAL EXAM PRECAUTIONS & VALUATION RUBRIC SECRETS

• **Alert:** Remember that S-waves cannot travel through liquids; their complete disappearance beyond 105 degrees from an earthquake epicenter provides definitive proof that the outer core is liquid.

• **Alert:** The P-wave shadow zone is an annular belt between 105 degrees and 145 degrees, whereas the S-wave shadow zone is massive, encompassing the entire region beyond 105 degrees (over 40 percent of Earth surface).

• **Alert:** The asthenosphere is located in the upper mantle (depth 80 to 200 km) and is in a semi-molten, ductile plastic state, serving as the main source of volcanic magma.

HIGH-YIELD CENTUM QUESTIONS & STEP-BY-STEP EXPLANATIONS

1. Differentiate between direct and indirect sources of information about the interior of the Earth with two examples of each. [Score: 2 Marks]

Answer & Key Valuation Steps:

• **Direct Sources:** Physical materials directly accessible to scientists, such as deep mining samples, deep ocean drilling cores (e.g. Kola deep borehole), and volcanic lavas. **[1 Mark]**

• **Indirect Sources:** Deductions based on properties of matter such as temperature and pressure increases with depth, gravity anomalies, magnetic surveys, and seismic wave trajectories. **[1 Mark]**

2. What is an earthquake shadow zone? Why does it occur? [Score: 2 Marks]

Answer & Key Valuation Steps:

• **Definition:** A specific area on the Earth's surface where seismographs do not detect earthquake waves from a particular epicenter. **[1 Mark]**

• **Causal Mechanism:** Caused by the refraction of P-waves as they enter the denser core and the total absorption and blockage of transverse S-waves by the liquid outer core. **[1 Mark]**

3. Compare P-waves and S-waves on the basis of particle motion, propagation velocity, and media of travel. [Score: 3 Marks]

Answer & Key Valuation Steps:

• **P-Waves (Primary):** Longitudinal compressional waves where particles vibrate parallel to the direction of wave propagation; fastest waves; travel through solids, liquids, and gases. **[1.5 Marks]**

• **S-Waves (Secondary):** Transverse shear waves where particles vibrate perpendicular to the direction of propagation creating crests and troughs; slower than P-waves; travel strictly through solid materials only. **[1.5 Marks]**

4. Explain the characteristics of the Earth's Mantle and describe the nature of the Asthenosphere. [Score: 3 Marks]

Answer & Key Valuation Steps:

• **Mantle Extent and Composition:** Extends from the Moho discontinuity to a depth of 2900 km, comprising about 84 percent of Earth's volume, composed primarily of dense peridotite rock rich in iron and magnesium. **[1 Mark]**

• **Asthenosphere Location:** The upper portion of the mantle extending between 80 km and 200 km depth, situated just below the rigid lithosphere. **[1 Mark]**

• **Physical State and Function:** Weak, semi-molten ductile plastic layer that allows tectonic plates to glide over it and supplies magma to volcanoes. **[1 Mark]**

5. Describe the three concentric layers of the Earth (Crust, Mantle, Core) along with their major boundary discontinuities. [Score: 4 Marks]

Answer & Key Valuation Steps:

• **Crust:** Outermost brittle layer; oceanic crust is thin (about 5 km, basaltic, density 2.7 g cm^{-3}), continental crust is thicker (about 30 km, granitic, density 3.0 g cm^{-3}). Separated from mantle by the Mohorovicic discontinuity. **[1.5 Marks]**

• **Mantle:** Extends to 2900 km depth; upper mantle contains ductile asthenosphere. Boundary between mantle and outer core at 2900 km is marked by the Gutenberg discontinuity. **[1.5 Marks]**

• **Core:** Heavy metallic core composed of Nickel and Iron (Nife); outer core (2900 to 5100 km) is liquid, while inner core (5100 to 6378 km) is solid under extreme pressure; separated by Lehmann discontinuity. **[1 Mark]**

6. Classify intrusive volcanic landforms into batholiths, laccoliths, dykes, and sills with their structural characteristics. [Score: 4 Marks]

Answer & Key Valuation Steps:

• **Batholiths:** Massive, deep-seated granitic rock bodies cooling deep in the crust, forming large dome-shaped granitic batholith cores of mountains when uncovered by erosion. **[1 Mark]**

• **Laccoliths:** Large dome-shaped intrusive bodies with a level base and connected by a pipe-like conduit from below, arching overlying strata into blisters. **[1 Mark]**

• **Sills and Sheets:** Concordant horizontal sheets of solidified magma intruded along bedding planes between rock layers; thin sheets are called sheets, thick sheets are called sills. **[1 Mark]**

• **Dykes:** Discordant wall-like vertical or near-vertical magmatic intrusions cutting across existing rock layers and fissures. **[1 Mark]**